Research Statement

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My research focuses on differential geometry, and on its intersection with smooth dynamical systems. More precisely, my work has advanced in two broad directions:

(i) Pseudo-Riemannian geometry, which is a generalization of the mathematical language of general relativity dealing with metric tensors which are not necessarily positive-definite, but instead only nondegenerate. Some results can be generalized from the positive-definite setting to the indefinite one, and some cannot. There are questions which are trivial or do not make sense in the Riemannian setting, but are open and unexplored otherwise, and results relating topology and geometry of indefinite metrics are generally rare—I'm largely interested in these last two types of problems.

(ii) Geometry of magnetic systems, which provide a mathematical formalism for studying the motions of particles subject to the action of a magnetic field. While the subject has received wide attention since its inception in the 1960s, especially regarding dynamical aspects of the associated magnetic flows and its orbits, many recent developments focus on more geometric aspects leading to rigidity results. I'm interested in such results and their relations with various new notions of magnetic curvature.

Here is a brief summary of my contributions to both (i) and (ii). They were obtained in joint work with a number of collaborators, all of which are properly mentioned later.

(i) We have established the existence of compact *ECS manifolds* (that is, those with parallel Weyl curvature, which are not locally symmetric or conformally flat) in all dimensions $n \geq 5$ and realizing all possible indefinite metric signatures [30, 31]. We also prove that non-locally-homogeneous compact rank-one ECS manifolds are (up to a double covering) all bundles over $\mathbb{S}^1$ [29], and that four-dimensional compact ECS manifolds, if they exist, must be of rank-two [28, 32]. Independently, I have proved that three-dimensional ECS manifolds (i.e., those with nonzero parallel Cotton tensor) cannot be compact [68].

(ii) We establish E. Hopf's theorem for magnetic systems without conjugate points, and characterize nontrivial flat magnetic systems as being Kähler [11]. Next, we establish marked length spectrum rigidity for Anosov magnetic systems on closed surfaces [10]. Further exploring the meaning of magnetic curvature, I wrote down magnetic versions of the classical submanifold compatibility equations [69]. We then prove, in the real-analytic category, that negatively-curved magnetic systems on closed manifolds admit only finitely many closed totally magnetic hypersurfaces, unless the magnetic field is trivial and the metric is hyperbolic [57]. Finally, we study the magnetic curvature of left-invariant magnetic systems on Lie groups [63].

In Sections 1 and 2, respectively, I will present the results mentioned above in more detail and give them some background and context; specific projects are discussed in subsections. I will also list questions which are unanswered and left for possible future work.

1 On pseudo-Riemannian manifolds

A *pseudo-Riemannian manifold* is a pair (M, g), where M is a smooth manifold and g is a nondegenerate twice-covariant symmetric tensor field on M. The fact that g is nondegenerate allows for most concepts of classical Riemannian geometry to carry through: isometries, Levi-Civita connection, geodesics, Killing vector fields, and curvature tensors all make sense without changes for pseudo-Riemannian manifolds. However, most analytic techniques in Riemannian geometry are no longer applicable, as they rely on positive-definiteness of g—for instance, the Laplacian operator for indefinite metrics is not elliptic.

Notably, spacetimes can be modeled as certain Lorentzian manifolds (i.e., those for which the negative index of g equals 1), and geometric objects associated with a spacetime have clear physical meanings: the metric is interpreted as the gravitational potential, the Levi-Civita connection becomes the gravitational field, while its curvature (absent in the setting of special relativity) describes what remains of gravity after all tidal forces are removed.

1.1 Parallel Weyl curvature and ECS manifolds (joint with A. Derdzinski)

The natural representation of the pseudo-orthogonal group $\mathrm{SO}(p,q)$ on the space of algebraic curvature tensors of a pseudo-Riemannian manifold (M,g) of signature (p,q) and dimension $n \geq 3$ allows us to write its Riemann curvature tensor as a sum $R = S + E + W$ of irreducible components: the *scalar part* S, the *Einstein part* E, and the *Weyl curvature tensor* W. The scalar and Einstein parts are built from the Ricci tensor, while W encodes all the information otherwise lost by tracing (which is none when $n = 3$).

The Weyl tensor has several well-known properties which illuminate its role in geometry, with the most important one being its conformal invariance. It can also be seen as the "Ricci-traceless" part of R and, when (M,g) is a spacetime, the part of the curvature which is not constrained by Einstein's field equations. For $n \geq 4$ we summarize in Table 1 the meaning of some zeroth and first order conditions imposed on S, E, and W; the case $n = 3$ is discussed in §1.2. Additional linear conditions include *harmonic curvature* and *harmonic Weyl curvature*: two topics in geometry with a prohibitively extensive literature.

Condition	Meaning	Condition	Meaning
$R = 0$	flatness	$\nabla R = 0$	local symmetry
$S = 0$	vanishing scalar curvature	$\nabla S = 0$	constant scalar curvature
$E = 0$	the Einstein condition: $\mathrm{Ric} = \lambda \mathrm{g}$	$\nabla E = 0$	local product of Einstein manifolds
$W = 0$	conformal flatness	$\nabla W = 0$	conformal symmetry

Table 1: The natural linear conditions imposed on irreducible components of the curvature tensor, in the sense of A. Besse [13].

Definition 1.1. A pseudo-Riemannian manifold of dimension at least 4 is called *conformally symmetric* if $\nabla W = 0$, and *essentially conformally symmetric (ECS)* if, in addition, $W \neq 0$ and $\nabla R \neq 0$.

Conformally symmetric manifolds were first considered in the early 1960s by M. C. Chaki and B. Gupta [19], and subsequently by T. Adati and T. Miyazawa [5]. The reason conformally symmetric manifolds have not received any attention in the setting of positive-definite metrics is a classical 1977 result by W. Roter, stating that a conformally symmetric Riemannian manifold is necessarily locally symmetric ($\nabla R = 0$) or conformally flat ($W = 0$) [24]. In other words, the nature of ECS manifolds is *strictly* pseudo-Riemannian, and W. Roter has also shown that ECS manifolds of all dimensions $n \geq 4$ and arbitrary indefinite metric signatures do exist [65]. The condition $\nabla W = 0$ has also found its place in the study of conformal transformations of pseudo-Riemannian manifolds, as illustrated by M. Cahen and Y. Kerbrat [17].

The local structure of ECS manifolds was described by A. Derdzinski and W. Roter [25, 26], in terms of a certain geometric invariant called the *rank*, always equal to 1 or 2: namely, it is the rank of a specific null parallel distribution $\mathcal{D}$ introduced by Z. Olszak [60]. With this in place, the next step is to understand the relation between the condition $\nabla W = 0$ and the topology of the manifold. Namely, our main goal is to understand the global structure of compact ECS manifolds.

Question 1. Do compact ECS manifolds exist? If so, must they be geodesically complete?

While A. Derdzinski and W. Roter constructed in [27] compact rank-one examples in dimensions of the form $n = 3k + 2$, with integers $k \geq 1$, many questions remained open. The first such question is whether compact ECS manifolds also exist in dimensions not of the form $3k + 2$. To answer it, we proved:

Theorem 1.2 (Derdzinski-T., '22 [31]). *There exist compact rank-one ECS manifolds of all dimensions $n \geq 5$ and all indefinite metric signatures, diffeomorphic to nontrivial torus bundles over $\mathbb{S}^1$, geodesically complete, and not locally homogeneous. Their local isometry types, in each fixed dimension and metric signature, form—in a natural sense—an infinite-dimensional moduli space.*

Our examples are built from what we call *rank-one ECS model manifolds*, or *standard ECS plane waves*: they represent all rank-one local isometry types and, as we later show [28], constitute the universal coverings of *generic* compact rank-one ECS manifolds (here, "genericity" refers to a certain algebraic condition imposed on the Weyl tensor.) More precisely, for such a model $(\widehat{M}, \widehat{g})$, we look for subgroups Γ of the isometry group $\mathrm{Iso}(\widehat{M}, \widehat{g})$ acting on $\widehat{M}$ freely and properly discontinuously in order to produce a compact quotient $M = \widehat{M}/\Gamma$.

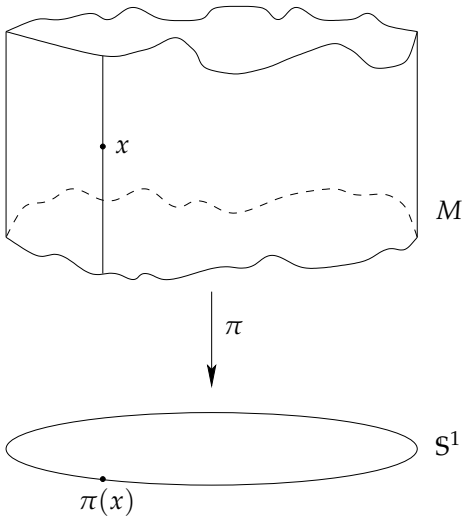


Figure 1: A compact rank-one ECS manifold M as the total space of a fiber bundle over a circle $\mathbb{S}^1$.

The fiber bundle structure exhibited by our examples is not accidental, but instead imposed by certain properties of the tangent foliation to $\mathcal{D}^\perp$, which consists of totally geodesic flat leaves; see Figure 1 for a simplified description. In general, we show:

Theorem 1.3 (Derdzinski–**Terek**, '22 [29]). *Every non-locally-homogeneous compact rank-one ECS manifold is, up to a double covering, the total space of a fiber bundle over $\mathbb{S}^1$ with fibers equal to the leaves of $\mathcal{D}^\perp$. The conclusion also holds for locally homogeneous manifolds, provided $\mathcal{D}^\perp$ has at least one compact leaf.*

The major part of our technical argument to establish this result consists in ensuring that $\mathcal{D}^\perp$—assumed to be transversely orientable—has at least one compact leaf L with a neighborhood saturated by compact leaves of $\mathcal{D}^\perp$ and diffeomorphically identified with the product $\mathbb{R} \times L$ so that $\mathcal{D}^\perp$ appears tangent to the L factor foliation. This ultimately allows us to apply a well-known result from foliation theory [59, p. 49]: a transversely orientable codimension-one foliation with compact leaves on a compact manifold M is tangent to the vertical distribution of a fiber bundle projection $M \to \mathbb{S}^1$.

In both Theorems 1.2 and 1.3, local homogeneity appeared as an obstacle. In addition, the four-dimensional case has eluded us thus far. Hence:

Question 2. Are there four-dimensional compact ECS manifolds? Can a compact rank-one ECS manifold be locally homogeneous?

Our approach to Question 2 relies on the genericity assumption mentioned after Theorem 1.2, as it justifies our focusing on the model manifolds $(\widehat{M}, \widehat{g})$. The isometry group $\mathrm{Iso}(\widehat{M}, \widehat{g})$ is explicitly computed to be a semidirect product between a certain subgroup of $\mathrm{Aff}(\mathbb{R}) \times \mathrm{O}(n)$ which interacts "correctly" with the Ricci and Weyl tensors of $(\widehat{M}, \widehat{g})$, and a Heisenberg group $\mathrm{H}(n)$ encoding essential information about the Killing equation in $(\widehat{M}, \widehat{g})$. With this in place, we show that a subgroup Γ of $\mathrm{Iso}(\widehat{M}, \widehat{g})$ acting on $\widehat{M}$ properly discontinuously with a compact quotient gives rise to a *lattice* Σ (that is, a finitely generated free Abelian group), on which Γ naturally acts. The local homogeneity assumption, the range of values for the rank of Σ, and the specific form of the spectrum of the elements of Γ acting linearly on Σ are incompatible:

Theorem 1.4 (Derdzinski–**Terek**, '23 [32, 28]). *Generic locally homogeneous rank-one ECS manifolds cannot be compact.*

This suggests that compact rank-one ECS manifolds might all be complete. In addition, for simple algebraic reasons, the genericity assumption is automatically satisfied when $\dim M = 4$. But it also was shown in [27] that outside of the locally homogeneous case four-dimensional models do not admit compact quotients. This, together with Theorem 1.4 above, leads to a partial answer to Question 2:

Theorem 1.5 (Derdzinski–**Terek**, '23 [28])**.** *Four-dimensional rank-one ECS manifolds are noncompact.*

Our complete focus on the rank-one case raises a natural question.

Question 3. Must all compact ECS manifolds be necessarily of rank-one?

Rank-two ECS manifolds are considerably more complicated to deal with than the rank-one ones. On one hand, rank-one ECS manifolds locally arise as *pp-wave manifolds*, used in general relativity as models for gravitational waves—as such, pp-wave manifolds have received a great deal of attention since their introduction by J. Ehlers and W. Kundt in the 1960s [35]. On the other hand, rank-two ECS manifolds locally arise as warped products of Riemann extensions (originally defined by E. M. Patterson and A. G. Walker [62]) of projectively flat connections on surfaces, with flat manifolds.

There is a very narrow class of rank-two ECS manifolds which may, however, be locally described within the class of pp-waves—this is the reason we eventually replaced the name "rank-one ECS model" with "standard ECS plane wave," to avoid referencing the rank. By modifying the quotient-construction previously applied to our standard ECS plane waves, we stumbled upon what turned out to be compact rank-two ECS manifolds:

Theorem 1.6 (Derdzinski–**Terek**, '23 [30])**.** *There exist compact rank-two ECS manifolds of all odd dimensions $n \geq 5$, including locally homogeneous ones, forming the total space of a nontrivial torus bundle over $\mathbb{S}^1$ with its fibers being the leaves of $\mathcal{D}^{\perp}$, all of them geodesically incomplete. In each fixed odd dimension, there is an infinite-dimensional moduli space of local-isometry types.*

These last examples turns out to complete our answers to both Question 1 and Question 3. The investigation of the global structure of rank-two ECS manifolds—focusing on answering Question 2—is left for future work.

1.2 Parallel Cotton tensor and pp-wave manifolds (joint with R. Sánchez Galán)

The Weyl tensor of any three-dimensional pseudo-Riemannian manifold (M, g) vanishes for purely algebraic reasons, and conformal flatness is controlled by its *Cotton tensor* instead—denoted by C and originally defined by É. Cotton in his study of three-dimensional conformal geometry [22]. Namely, (M, g) is conformally flat if and only if $\mathrm{C} = 0$. So, in general, the "conformal curvature tensor" measuring the deviation from conformal flatness, is really the *pair* (W, C) (it is well-known that C is determined by W in dimensions greater than three via the formula $(n-3)\mathrm{C} = (n-2)\mathrm{div}_{\mathrm{g}} W$ [13]). Motivated by this, E. García-Rio et. al. [18] proposed a natural extension of Definition 1.1 to the three-dimensional case:

Definition 1.7. A three-dimensional pseudo-Riemannian manifold is called *conformally symmetric* if $\nabla \mathrm{C} = 0$, and *essentially conformally symmetric* if, in addition, $\mathrm{C} \neq 0$.

Thus, "conformal symmetry" means "parallel conformal curvature tensor" in all dimensions. Still in [18], they described the local structure of three-dimensional ECS manifolds. As a consequence of this local structure theorem, the kernel of C in *any* three-dimensional ECS manifold (M, g) defines a rank-one null parallel distribution $\mathcal{D}$ on M, similar to the Olszak distribution. With this and techniques similar to the ones described in §1.1, I settled the three-dimensional version of Question 1:

Theorem 1.8 (**Terek** '23 [68])**.** *Every compact three-dimensional conformally symmetric pseudo-Riemannian manifold is necessarily conformally flat.*

In other words, there are no compact three-dimensional ECS manifolds. However, the relevance of the Cotton tensor is not restricted to three-dimensional manifolds: a gravity theory attempting to extend general relativity by replacing $\mathrm{Ric} - \mathrm{sg}/2$ and the energy-momentum tensor T in Einstein's field equations with the Cotton tensor and covariant derivatives of T, respectively, has been proposed by J. Harada in 2021, who also exhibited a family of Schwarzschild-type vacuum solutions to the new proposed equations [49, 51]. It is also worth noting that pp-wave spacetimes have also been investigated in the setting of Cotton gravity by M. Gürses et al. [47]. While Harada's theory has been the subject of many discussions and the target of some criticism [12, 21, 50, 67], interest in the geometry of the Cotton tensor has only

increased; for instance, some vanishing results for the Cotton tensor on compact manifolds have been obtained by L. F. Wang [72].

The parallel Cotton tensor condition, in particular, has also appeared in reasonably recent work: A. Ghosh proved that Kenmotsu manifolds with parallel Cotton tensor must be Bach-flat [39], and Riemannian 4-manifolds with parallel Cotton tensor have received attention from a number of authors [20, 38]. We thus consider the following natural question:

Question 4. Must a compact pseudo-Riemannian manifold of dimension $n \geq 4$ with parallel Cotton tensor have vanishing Cotton tensor?

The affirmative answer to Question 4 in the Riemannian signature is a trivial consequence of the divergence expression for C together with the following elementary general fact: for a compact Riemannian manifold (M, g), any TM-valued covariant tensor field on M with parallel divergence is necessarily divergence-free. On the other hand, Question 4 is particularly interesting in the four-dimensional Lorentzian signature—compact spacetimes always admit closed timelike curves, meaning that reasonable spacetime models for the universe should be noncompact.

In joint work with R. Sánchez Galán, I am investigating the geometry of higher-dimensional manifolds with parallel Cotton tensor. While the local structure of three-dimensional ECS manifolds does match the local structure of rank-one ECS manifolds in dimensions $n \geq 4$, nothing is known about it otherwise. The parallel distribution $\mathcal{D}$ is no longer guaranteed to be null, or of rank 1 or 2, but we have shown that $\dim \mathcal{D} \leq n-2$, with equality both implying that $\mathcal{D}$ is degenerate and allowing for a precise description of the algebraic type of the pair (g, C). This "maximal nullity" situation is always the case for pp-wave spacetimes and, in this setting, we have successfully generalized Theorem 1.8; we also consider other types of wave-type spacetimes, such as pure-radiation waves introduced by T. Leistner [54]. By means of products and a higher-signature version of pp-wave manifolds, we exhibit examples of pseudo-Riemannian manifolds of dimension $n \geq 4$ having nonzero parallel Cotton tensor realizing all possible indefinite metric signatures. Another interesting question regards what the possible holonomy groups of the connection induced on $\mathcal{D}$. Important tools for the study of such problems include the classical results of A. G. Walker [70, 71], providing special coordinates adapted to any given parallel degenerate distribution on a pseudo-Riemannian manifold.

2 On geometry of magnetic systems

Definition 2.1. A *magnetic system* on a smooth n-manifold M consists of a pair (g, σ), where g is a Riemannian metric and σ is a closed 2-form. The *Lorentz force operator* of (g, σ) is the skew-adjoint endomorphism $\mathbf{Y}$ of TM g-dual to σ.

Closedness of σ reflects that magnetic fields are divergence-free, and solutions of the *Landau-Hall equation*

$$\frac{\mathrm{D}\dot{\gamma}}{\mathrm{d}t}(t) = \mathbf{Y}_{\gamma(t)}(\dot{\gamma}(t)), \tag{LH}$$

called (g, σ)-*geodesics* (or, *magnetic geodesics*), model trajectories of particles in (M, g) subject to the action of the magnetic field represented by σ; this formalism goes back to D. V. Anosov and Y. G. Sinai [7] and V. I. Arnol'd [8] in the 1960s. The *magnetic flow* induced by (LH) is Hamiltonian, and leaves all s-sphere bundles $\Sigma_s \subseteq TM$ invariant. The dynamical properties of the restricted s-*magnetic flow* on Σ_s now strongly depends on the value of $s > 0$ as (LH) is in general not homogeneous, and one of the main tools for studying magnetic flows are (g, σ)-*Jacobi fields*, obtained as variational vector fields of variations of (g, σ)-geodesics and satisfying the linearization of (LH).

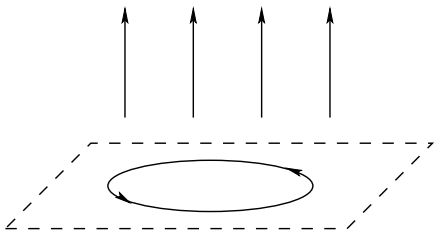


Figure 2: All magnetic geodesics for a uniform field $\sigma = \lambda \, \mathrm{d}x \wedge \mathrm{d}y$, i.e. with constant $\lambda \in (0, \infty)$, on the plane $(\mathbb{R}^2, \mathrm{d}x^2 + \mathrm{d}y^2)$ are closed. (It is one example of a *Zoll magnetic system*.)

In 2024, aiming to establish the existence of closed orbits of the s-magnetic flow for low values of $s > 0$, V. Assenza introduced in [9] a notion of *s-magnetic curvature operator* $\boldsymbol{M}_v^{\mathrm{g},\sigma,s} \colon v^\perp \to v^\perp$, which generalizes the well-known Jacobi operators $w \mapsto R(w,v)v$ appearing in the classical Jacobi equation. When v, w are orthonormal, one may consider the corresponding *s-magnetic sectional and Ricci curvatures*

$$\mathrm{Sec}^{(\mathrm{g},\sigma,s)}(v,w) = \mathrm{g}(\boldsymbol{M}_v^{(\mathrm{g},\sigma,s)}(w), w) \quad \text{and} \quad \mathrm{Ric}^{(\mathrm{g},\sigma,s)}(v) = \mathrm{tr}(\boldsymbol{M}_v^{(\mathrm{g},\sigma,s)}).$$

While the definition of $\boldsymbol{M}^{(\mathrm{g},\sigma,s)}$ is technical, it turns out that many results from Riemannian geometry can be extended to the magnetic setting with it. More details are given in the following sections.

2.1 E. Hopf's Theorem (joint with V. Assenza and J. Marshall Reber)

A classical result by E. Hopf [52] states that if a closed Riemannian surface has no conjugate points, then its total Gaussian curvature is nonpositive, and vanishes if and only if the metric is flat. This result has been generalized to higher dimensions by Green [43], with the scalar curvature replacing the Gaussian curvature.

Question 5. Does E. Hopf's (or more generally, Green's) theorem hold for magnetic systems without conjugate points? If so, what is the meaning of magnetic flatness?

In 1998, even without a notion of magnetic curvature operator, N. Gouda [42] extended Hopf and Green's results to *uniform* magnetic systems (i.e., those for which $\nabla\sigma = 0$). With the correct language of magnetic curvature at our disposal, we were able to fully adapt Green's original argument to the magnetic setting. Specifically, we constructed magnetic versions of the so-called *Green bundles*, roughly as follows:

Step 1: With $\mathbf{X}^{\mathrm{g},\sigma}$ denoting the infinitesimal generator of the s-magnetic flow $\Phi^{(\mathrm{g},\sigma,s)}$ and $Q = T\Sigma_s/\mathbf{X}^{\mathrm{g},\sigma} \to \Sigma_s$ the corresponding symplectic vector bundle, the vertical distribution of the fibration $T\Sigma_s \to \Sigma_s$ projects onto a subbundle $\mathcal{V}$ of Q, which can be moved around by means of the quotient-flow on Q induced by the derivative of $\Phi^{(\mathrm{g},\sigma,s)}$, giving rise to a family $E(t)$ of Lagrangian subbundles of Q.

Step 2: The absence of magnetic conjugate points translates into a transversality condition on $E(t)$, which allows us to identify each $E(t)_v$ with the graph of a self-adjoint endomorphism of $v^\perp$; with this, we could establish that the limits $E^\pm = \lim_{t\to\pm\infty} E(t)$ exist and are Lagrangian subbundles of Q as well.

Step 3: Using the self-adjoint endomorphisms representing the limits $E^\pm$, we produced solutions to a certain Riccati differential equation involving the s-magnetic Ricci curvature. Integrating such equation, the conclusion follows.

The most challenging technical aspect to overcome when carrying out this entire strategy was the fact that, while in the Riemannian setting perpendicular parts of Jacobi fields are again Jacobi fields, this is no longer the case in the magnetic setting. Instead, perpendicular parts of magnetic Jacobi fields satisfy a differential equation involving the s-magnetic curvature operator, which is the direct analogue of the Riemannian Jacobi equation (and hence fitting into the general framework of Morse-Sturm systems), but the Levi-Civita covariant derivative is replaced with a *magnetic covariant derivative operator* along the magnetic geodesic in question.

Theorem 2.2 (Assenza–Marshall Reber–**Terek**, '24 [11])**.** *Let (g,σ) be a magnetic system on a closed manifold M, and $s > 0$ be such that the s-magnetic flow is without conjugate points. Then*

$$\int_{UM} \mathrm{Ric}^{(\mathrm{g},\sigma,s)}(v)\, \mathrm{d}\mu^{\mathrm{g}}(v) \leq 0,$$

with equality if and only if $\boldsymbol{M}^{(\mathrm{g},\sigma,s)} = 0$; here, μ^{g} is the Liouville measure on the unit-tangent bundle UM of (M,g).

As another application of the existence of magnetic Green bundles, we were also able to establish the hyperbolic behavior of negatively curved magnetic flows, generalizing other results of N. Gouda [41] for uniform fields, and of M. P. Wojkowski [73].

Theorem 2.3 (Assenza–Marshall Reber–**Terek**, '24 [11]). *Whenever (g, σ) is a magnetic system on a closed manifold and $s > 0$ has $\mathrm{Sec}^{(\mathrm{g},\sigma,s)} < 0$, the s-magnetic flow is Anosov.*

As a consequence of E. Hopf's original result and the Gauss-Bonnet theorem, the only metrics without conjugate points on the torus $\mathbb{T}^2$ are the flat ones. Without a higher-dimension version of the Gauss-Bonnet theorem, the question of whether metrics without conjugate points on $\mathbb{T}^n$ must be flat remained open until 1994, when D. Burago and S. Ivanov [15] answered it in the affirmative with a completely different array of techniques. We complete our answer to Question 5 by obtaining a description of the nontrivial magnetically flat systems; in fact, we consider the more general condition of constant s-magnetic sectional curvature.

Theorem 2.4 (Assenza–Marshall Reber–**Terek**, '24 [11]). *Let (g, σ) be a nontrivial magnetic system on a smooth n-dimensional manifold M, and let $s > 0$ be such that $\mathrm{Sec}^{(\mathrm{g},\sigma,s)} \equiv c$, for some $c \in \mathbb{R}$. Then $\nabla\sigma = 0$, σ is nowhere-vanishing, and one of the following must hold:*

(i) *$n = 2$, (M, g) is an oriented surface with constant Gaussian curvature $(c - \|\mathbf{Y}\|^2)/s^2$, and $\sigma = \|\mathbf{Y}\|\nu_{\mathrm{g}}$, where ν_{g} is the Riemannian area form of g.*

(ii) *$n \geq 4$, $c = 0$, and $J = \|\mathbf{Y}\|^{-1}\mathbf{Y}$ is a complex structure for which (M, g) becomes a Kähler manifold of constant holomorphic sectional curvature $-\|\mathbf{Y}\|^2/s^2$.*

While it may seem that the story ends with Theorem 2.4, it is natural to consider further geometric conditions—inspired by Riemannian geometry—to be imposed on the magnetic curvature, and attempt to classify the corresponding magnetic systems. Here are two examples.

Question 6. Which systems (g, σ) are *conformally s-magnetic flat*, that is, admit functions $\alpha, \beta > 0$ for which $(\alpha\mathrm{g}, \beta\sigma)$ is s-magnetically flat? Can such condition be characterized by the vanishing of some would-be s-magnetic Weyl tensor $\boldsymbol{W}^{(\mathrm{g},\sigma,s)}$?

Question 7. Which systems (g, σ) are *s-magnetically Einstein*, that is, have $\mathrm{Ric}^{(\mathrm{g},\sigma,s)} \equiv c$ a constant?

Conformally s-magnetic flat systems are evidently closely related to conformally Kähler metrics and, in the same way that s-magnetically flat systems have $\nabla\sigma = 0$, the s-magnetically Einstein ones are readily seen to have $\mathrm{div}_{\mathrm{g}}\sigma = 0$ (thus σ, being also closed, is harmonic). Further exploration is left for future work.

2.2 Magnetic marked length spectrum rigidity for Anosov surfaces (joint with V. Assenza, J. De Simoi, and J. Marshall Reber)

It is well-known that, whenever (M, g) is a negatively-curved Riemannian manifold, every nontrivial free homotopy class α of loops $\mathbb{S}^1 \to M$ contains a unique closed geodesic γ_α. Let $\mathcal{C}$ denote the set of all such homotopy classes and L_{g} be the length function associated with g.

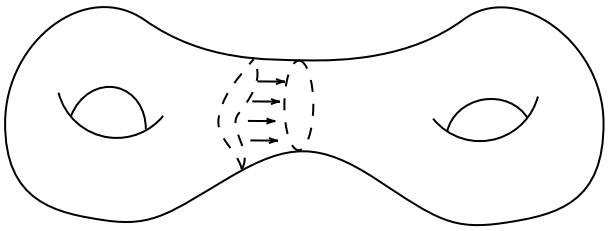


Figure 3: A closed curve in a nontrivial free homotopy class in a hyperbolic surface being deformed into the corresponding closed geodesic representative, which has minimal length.

Definition 2.5. The *marked length spectrum* of the metric g is the function $\mathrm{MLS}(\mathrm{g})\colon \mathcal{C} \to [0, \infty)$ given by $\mathrm{MLS}(\mathrm{g})(\alpha) = L_{\mathrm{g}}(\gamma_\alpha)$.

The *marked length spectrum conjecture* [16] states that MLS is injective on metrics, that is, two negatively curved metrics on a closed manifold having the same marked length spectrum must be isometric. For surfaces, this conjecture has been independently proved by J.-P. Otal [61] and C. B. Croke [23] and, more recently, generalized for metrics with Anosov geodesic flow by C. Guillarmou, T. Lefreuve, and G. P. Paternain [45]. Results in higher dimensions are more scarce: A. Katok [53] proved that the restriction of MLS to single conformal classes is injective, while U. Hamenstädt [48] showed that the conjecture is true when one of metrics in question is assumed to be locally symmetric.

This conjecture also makes sense in the magnetic setting. More precisely, whenever a magnetic system (g,σ) on a smooth manifold and $s > 0$ is such that $\mathrm{Sec}^{(g,\sigma,s)} < 0$, the s-magnetic flow is Anosov and therefore for every $\alpha \in \mathcal{C}$ there is a unique closed (g,σ)-geodesic $\beta_\alpha \in \alpha$ with constant speed s; we assume from here onwards that $s = 1$, for simplicity. We may now talk about the *magnetic marked length spectrum* of (g,σ), which is the function $\mathrm{MLS}(g,\sigma)\colon \mathcal{C} \to [0,\infty)$ defined by $\mathrm{MLS}(g,\sigma)(\alpha) = L_g(\beta_\alpha)$.

Question 8. Is the magnetic marked length spectrum mapping injective on magnetic systems? In other words, given two negatively-curved (or, more generally, Anosov) magnetic systems on a closed manifold M having $\mathrm{MLS}(g,\sigma) = \mathrm{MLS}(g,\sigma)$, is there a diffeomorphism $f\colon M \to M$ with $f^*(g',\sigma') = (g,\sigma)$?

At this level of generality, little progress has been made towards answering Question 8. S. Grognet [44] considered this question for surfaces, under the assumption that the magnetic flow in question is conjugate to the geodesic flow of a negatively curved metric, while J. Marshall Reber [56] established a deformative version of magnetic marked length spectrum rigidity for surfaces (originally done in the Riemannian setting by V. Guillemin and D. Kazhdan [46]). Moreover, by a result of Epstein-Gornet-Mast [36], on compact quotients of Heisenberg groups with invariant magnetic systems the magnetic marked length spectrum determines the Riemannian metric.

Focusing only on the case of surfaces, we provide an answer to Question 8:

Theorem 2.6 (Assenza–De Simoi–Marshall Reber–**Terek**, '24 [10])**.** *Let Σ be a closed surface and (g,σ) and (g',σ') be two Anosov magnetic systems such that*

$$\mathrm{MLS}(g,\sigma) = \mathrm{MLS}(g',\sigma'), \quad \mathrm{Area}(\Sigma,g) = \mathrm{Area}(\Sigma,g'), \quad \textit{and} \quad \int_\Sigma \sigma = \int_\Sigma \sigma'.$$

Then, there is a diffeomorphism $f\colon \Sigma \to \Sigma$, isotopic to the identity, such that $f^*(g',\sigma') = (g,\sigma)$.

We also show by means of an example that the assumption that $[\sigma] = [\sigma']$ in cohomology, phrased as the equality between integrals, cannot be removed. This reflects the fact that nontrivial magnetic flows are not reversible, and in particular one may have $\mathrm{MLS}(g,\sigma)(\alpha) \neq \mathrm{MLS}(g,\sigma)(-\alpha)$.

The big picture of our argument follows the one in [45], with a large number of recent results serving as crucial ingredients. Equality of the magnetic marked length spectra implies, by Livshits' theorem, that the magnetic flows of (g,σ) and (g',σ') are C^0-conjugate. Then, the smooth ridigity theorem for three-dimensional volume-preserving Anosov flows by A. Gogolev and F. Rodriguez-Hertz [40] implies that the magnetic flows are in fact C^∞-conjugate. Under these conditions, J. Echevarría Cuesta has shown in [33] that g' must be conformal to g. It would then remain to argue that the conformal factor is trivial, and this was the most challenging part of this project. In the geodesic case, this ultimately follows from closed geodesics being minimizers, in their free homotopy class, of the length functional. We achieved this in the exact magnetic setting by using that magnetic geodesics are similarly minimizers of a certain functional, suggestively called the *magnetic length* functional; the non-exact case is treated separately, by using that the pullbacks of σ and σ' to the unit-tangent bundle of Σ are themselves exact.

Question 8, in higher dimensions and with volume replacing area, remains very challenging and open.

2.3 Other projects on magnetic geometry

2.3.1 Submanifold compatibility equations

Submanifold theory, one of the broadest topics within Riemannian geometry, consists of studying the geometry of submanifolds in terms of the geometry of an ambient manifold containing them. This is all carried out by using a certain tensor, called the *second fundamental form*, which essentially encodes the part of the curvature external to the submanifold. As it turns out, this second fundamental form must satisfy a number of so-called compatibility equations, due to Gauss, Codazzi, and Ricci, that guide the entire theory.

A special class of submanifolds consists of the *totally geodesic* ones, that is, those second fundamental form vanishes. They are also characterized by the property that geodesics starting tangent to it, remain in it for small enough time. In the magnetic setting, one may of course say that a submanifold is *totally magnetic*

if magnetic geodesics satisfy this last property. In the first arXiv version of a recent work by Albers-Benedetti-Maier [6], among other things, they introduce a magnetic version of the second fundamental form, whose vanishing precisely characterizes the property of being totally magnetic. Further extending their definition of magnetic second fundamental form, I wrote down in [69] magnetic versions of the compatibility equations mentioned above. The resulting formulas, while not very friendly, might prove useful in extending classical results from submanifold theory to the magnetic setting.

2.3.2 Finiteness of totally magnetic hypersurfaces (joint with J. Marshall Reber)

Constant-curvature manifolds play a central role in Riemannian geometry, and many results provide different characterizations of this condition. In a recent work of Filip-Fisher-Lowe [37], it is proved that if a closed real-analytic negatively-curved Riemannian manifold (M, g) has infinitely many closed totally geodesic hypersurfaces, then (M, g) is arithmetic hyperbolic. Replacing g with a magnetic system (g, σ) and "totally geodesic" with "totally magnetic," it now makes sense to ask what the resulting condition on M and (g, σ) must be. We find that the situation remains incredibly rigid in the magnetic setting.

Theorem 2.7 (Marshall Reber–**Terek**, '26 [57])**.** *Let (g, σ) be a real-analytic magnetic system on a closed and connected real-analytic manifold M of dimension at least 3. If there is $s > 0$ such that $\mathrm{Sec}^{(\mathrm{g},\sigma,s)} < 0$ and M contains infinitely many closed totally magnetic hypersurfaces, then $\sigma = 0$ and (M, g) is arithmetic hyperbolic.*

While a large part of the argument given in [37] is topological and carried out to our case without any challenges, we dealt with the remaining aspects of it in considerably more generality, introducing dynamical versions of the second fundamental form—incorporating [6, 69]—and exponential mappings which are potentially of independent interest. In the process, we have also established a dynamical version of Cartan's axiom of the k-planes, whose conclusion entails constancy of the underlying sectional curvature and also implies that the flow considered is a smooth time-change of the corresponding geodesic flow.

2.3.3 The curvature of left-invariant magnetic systems (joint with S. Rankin and D. Weed)

The simplest examples of homogeneous spaces in Riemannian geometry are Lie groups equipped with left-invariant metrics; a classical paper by J. Milnor [58] contains a number of results relating curvature and algebraic properties. While integrability of left-invariant magnetic flows on Lie groups has received some attention in the literature [14, 34, 55], their curvature had not been investigated. In our preprint [63], aiming to extend Milnor's results to the magnetic setting, we compute the magnetic curvature of left-invariant systems, and establish the following:

(i) If (g, σ) is a bi-invariant magnetic system and $s > 0$, then $\mathrm{Sec}^{(\mathrm{g},\sigma,s)} \geq 0$, with equality if and only if $\sigma = 0$ and G is Abelian.

(ii) There is $s > 0$ such that $\mathrm{Sec}^{(\mathrm{g},\sigma,s)} = 0$ if and only if $\sigma = 0$ and $\mathfrak{g} = \mathfrak{u} \oplus \mathfrak{b}$ with $\mathfrak{u}$ a commutative ideal, $\mathfrak{b}$ an Abelian subalgebra, and ad_b skew-adjoint for all $b \in \mathfrak{b}$, or if $\sigma \neq 0$ and (G, g) equipped with the complex structure $J = \|\mathbf{Y}\|^{-1}\mathbf{Y}$ is holomorphically isometric to a complex hyperbolic space.

(iii) If there is $s > 0$ such that $\mathrm{Sec}^{(\mathrm{g},\sigma,s)} = 0$, then G is solvable. If, in addition, G is unimodular, then $\sigma = 0$ and g is flat.

Without a magnetic version of the Cartan-Hadamard theorem available, and hence without a classification of negatively-curved homogeneous magnetic systems, one question from this project is left open.

Question 9. If a connected Lie group G admits a nontrivial left-invariant magnetic system (g, σ) and $s > 0$ such that $\mathrm{Sec}^{(\mathrm{g},\sigma,s)} < 0$, must G be solvable?

2.3.4 Rauch's comparison theorem for magnetic systems (joint with F. Wilhelm)

One of the most fundamental comparison theorems in Riemannian geometry—concerning estimates of norms of Jacobi fields—is due to H. E. Rauch [64]. A number of generalizations of Rauch's theorem for Kähler magnetic systems has been obtained by T. Adachi and his students [1, 2, 3, 4, 66]; such results, however, predate the notion of magnetic curvature operator. In ongoing work with F. Wilhelm, using that perpendicular parts of (g, σ)-Jacobi fields satisfy a differential equation fitting into the framework of abstract Morse-Sturm systems, we are able to apply Riccati comparison results to extend the magnetic version of Rauch's theorem beyond the Kähler case.

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